

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)	Force down on magnet or the balance (due to positive) (1) N3 or equal and opposite mentioned (1)		2		2		2
		(ii)	Right side is N (on diagram or stated) (1) [F] LHR (1)	1	1		2		2
		(iii)	Forces are horizontal OR forces are equal and opposite Accept forces are outwards and inwards NOT wire not perpendicular to field		1		1		1
	(b)	(i)	12.0 [$\times 10^{-3}$ N] (1) 3.65 [g] (1) 2 points plotted correctly within half square (2) ecf on 12.0 [$\times 10^{-3}$ N] 1 point plotted correctly within half square (1) 0 points plotted correctly within half square (0)		4		4	4	4
		(ii)	Ignored the last 2 points (1) Since [partly] outside uniform field (1) Award the first 2 marks for: Has only used the points in the uniform field Line good [with same number above and below] (1)			3	3		3

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		(iii)	Conversion of cm (1) Correct method for gradient (1) Gradient = 0.583 (1) Gradient equated to BI (1) Field = 0.115 T unit mark (accept 0.11 to 0.12 to 2sf or 3sf) (1) Note 1.1 T to 1.2 T obtains 4 marks Alternative: Obtaining point on line – don't award for point more than $\frac{1}{2}$ square from line, don't award if point has a length for XY less than 3 cm (1) Converting cm to m (1) Rearranging equation i.e. $B = \frac{F}{Il}$ (1) Substituting (1) Field = 0.115 T unit mark (accept 0.11 to 0.12 to 2sf or 3sf) (1) Note 1.1 T to 1.2 T obtains 4 marks			5	5	5	5
			Question 4 total	1	8	8	17	9	17